

Personalized Debiasing for Sequential Recommendation

Addressing Popularity & Temporal Trending Bias with User Identity Scores

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The Problem: Why Personalized Debiasing?

Popularity Bias

Models over-recommend globally popular items because they dominate training data. Niche users suffer the most, almost all their recs are popular items they don't want.

Temporal Trending Bias

Currently trending items get pushed to ALL users, even those with stable long-term interests. Not every user is a trend-follower.

Uniform Debiasing Fails

Existing IPS methods apply the same correction to every user. A Niche explorer and a Blockbuster fan receive identical debiasing that is clearly suboptimal.



Key insight: Different users need different debiasing strengths — debiasing must be personalized.

Personalized Dual-Bias Debiasing Framework

$$L = \frac{1}{e_{\text{pop}}(i)^{\alpha(u)} \times e_{\text{trend}}(i, t)^{1-\beta(u)}} \times (-\log P(i, t))$$

Explorer Score $\alpha(u)$

Normalized Shannon entropy over genre distribution.

$\alpha \approx 1.0 \rightarrow$ diverse explorer $\rightarrow$ strong popularity correction

$\alpha \approx 0.0 \rightarrow$ loyal user $\rightarrow$ minimal correction

Trend-Follower Score $\beta(u)$

Average trending exposure across user's history.

$\beta \approx 0.8 \rightarrow$ trend-follower $\rightarrow$ less trending correction

$\beta \approx 0.1 \rightarrow$ stable-interest $\rightarrow$ strong trending correction



Model-agnostic (GRU4Rec / SASRec)



Training-time only — zero inference overhead



Formula-based & MLP-learned

Experimental Setup

Dataset: MovieLens-1M

~1M ratings | 6K users | 4K movies | 18 genres

Implicit feedback: ratings ≥ 4.0 as positive

Min 5 interactions per user

Per-user temporal holdout split

GRU4Rec Configuration

Embed: 128 | Hidden: 256 | Layers: 1

LR: 0.001 | Weight Decay: 1e-4

Batch: 512 | Epochs: 50 | Patience: 10

Trend window $W = 30$ days | Max weight: 5.0

Models Compared

Model	Description
Popularity Baseline	Recommends most globally popular items. Non-personalized.
GRU4Rec CE	Standard GRU4Rec with cross-entropy loss. No debiasing.
GRU4Rec + IPS	Uniform IPS reweighting — same correction for all users.
Ours (Formula)	Personalized debiasing with formula-based $\alpha(u)$ and $\beta(u)$.
Ours (Learned)	Personalized debiasing with MLP-learned $\alpha(u)$ and $\beta(u)$.

Evaluation Metrics

Ranking Quality

Recall@10: Is the ground-truth item in the top-10 list? Binary (0 or 1), averaged over all test users.

NDCG@10: Rewards placing the correct item higher. Rank #2 $\rightarrow$ 0.63, Rank #9 $\rightarrow$ 0.29.

Fairness & Diversity

Δ GAP: Rec avg popularity – profile avg popularity. Closer to 0 = fairer.

Coverage: Unique items recommended. Higher = more catalog explored.

Long-Tail Ratio: Fraction from bottom 80% by popularity. Higher = niche exposure.

Intuitive Example

Alice's next movie is "Interstellar". The model produces a top-10 list.

Recall@10: If "Interstellar" appears anywhere in top-10 $\rightarrow$ 1, else $\rightarrow$ 0.

NDCG@10: If it appears at rank #2 $\rightarrow$ score $\approx$ 0.63. At rank #9 $\rightarrow$ score $\approx$ 0.29. Not in top-10 $\rightarrow$ 0.

Δ GAP: If Alice's profile has avg popularity 0.003 and her recs average 0.005, Δ GAP = +0.002 (system pushes more popular items than she prefers).

Main Comparison Results

Method	Recall@10 ↑	NDCG@10 ↑	Δ GAP Niche ↓	Avg Pop ↓	Coverage ↑	Long-Tail ↑
Popularity	0.0414	0.0193	0.00290	0.00376	0.0326	0.0000
GRU4Rec CE	0.1255	0.0651	0.00079	0.00191	0.4662	0.0917
GRU4Rec + IPS	0.1221	0.0641	0.00084	0.00193	0.4639	0.0875
Ours (Formula)	0.1231	0.0638	0.00083	0.00193	0.4713	0.0928
Ours (Learned)	0.1273	0.0649	0.00082	0.00192	0.4653	0.0918

Key Takeaways

Formula-based wins on diversity

Highest coverage (0.4713) and long-tail ratio (0.0928) — best exposure fairness.

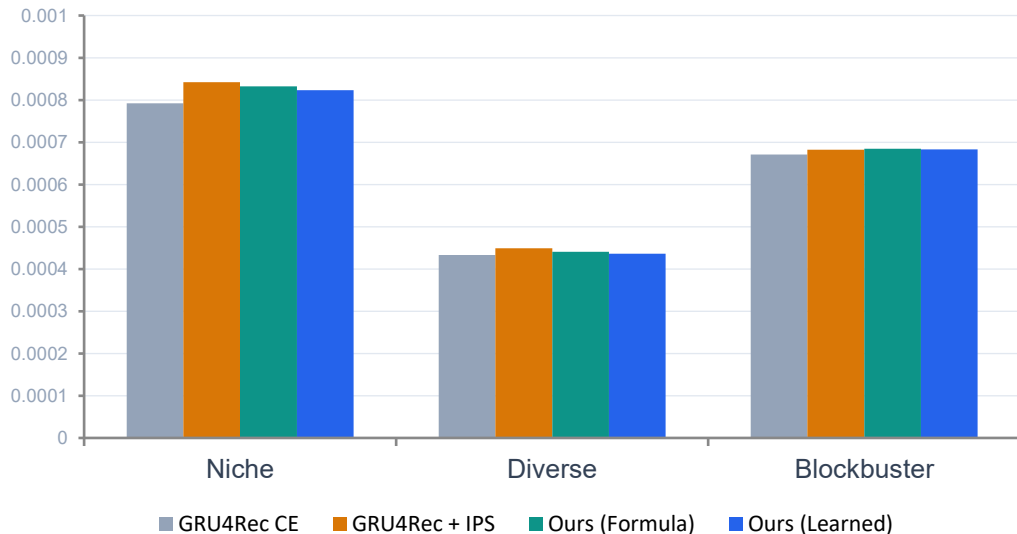
Learned wins on ranking

Best Recall@10 (0.1273) and NDCG@10 (0.0649) — ~3–4% gain over formula.

Both beat uniform IPS

Better Recall, Coverage, and Long-Tail than standard IPS — personalization matters.

Δ GAP Across User Groups



(Lower = fairer · Popularity baseline $\approx$ 0.0029, excluded for scale)

Differences are very small

All sequential methods sit at the 4th decimal. Practical differences are minimal.

Ours improves over uniform IPS

Both personalized variants achieve lower Δ GAP than IPS across all 3 user groups, supporting the personalization story.

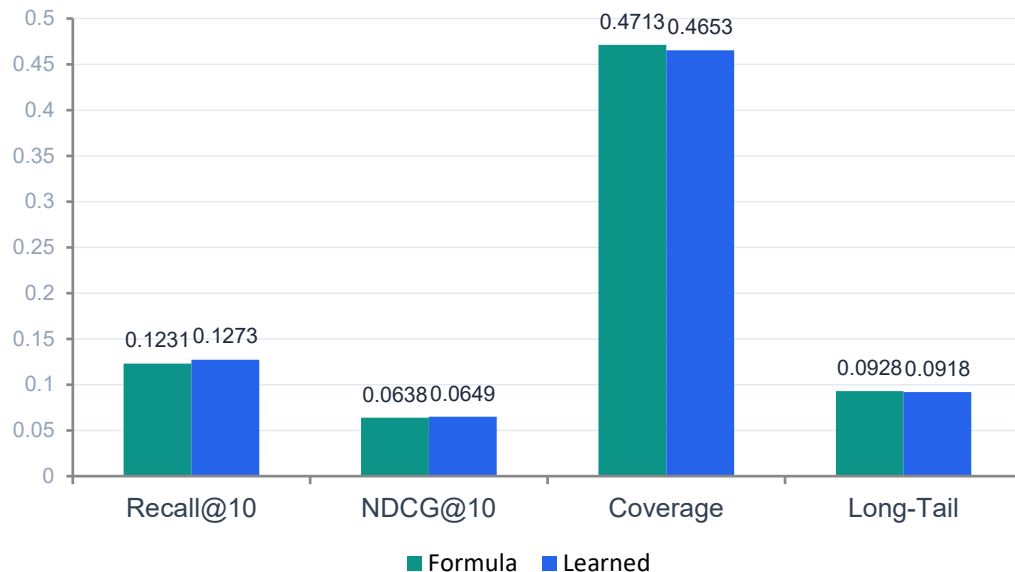
Plain CE is marginally best here

CE achieves slightly lower Δ GAP — but at the cost of Coverage and Long-Tail. Fairness has multiple dimensions.



Takeaway: Δ GAP differences are minimal; our gains show up in Coverage and Long-Tail.

Formula vs. Learned: The Trade-off



Learned MLP Wins

Better Recall@10 (+3.4%)

Better NDCG@10 (+1.7%)

Better Δ GAP Niche

Formula-Based Wins

Higher Coverage (+1.3%)

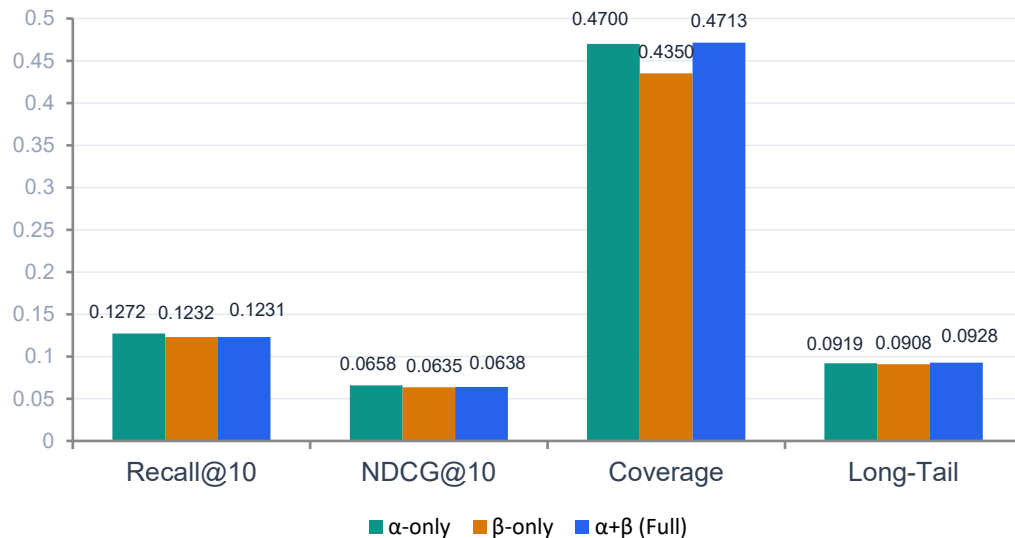
Higher Long-Tail Ratio (+1.1%)

More stable exposure control



Trade-off: Learned optimizes ranking; Formula provides more stable fairness and diversity control.

Ablation: α -only vs. β -only vs. $\alpha+\beta$



α dominates ranking

Popularity bias is the more dominant in MovieLens. Strong global popularity skew makes α the most impactful component.

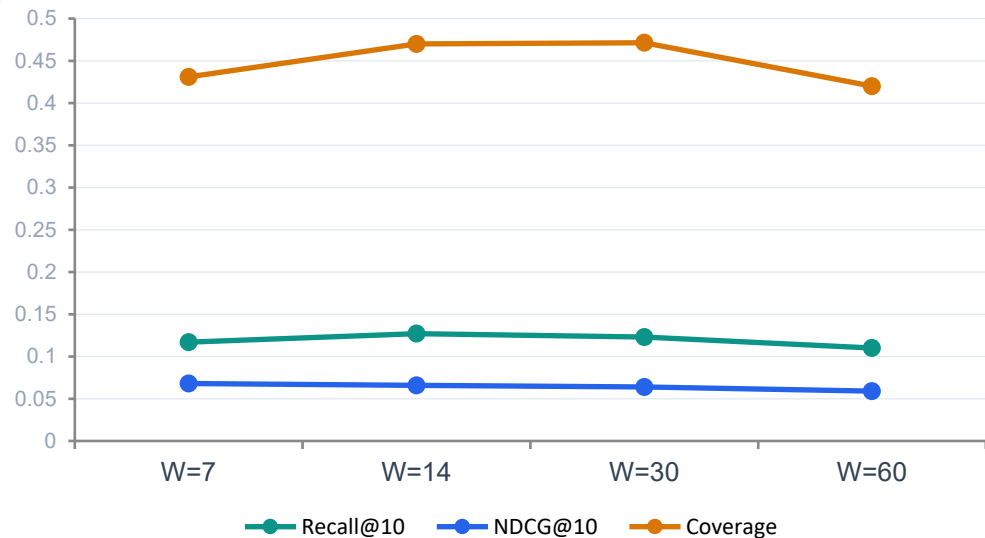
β contributes to fairness

β correction primarily addresses temporal trending bias. Any improvement in popularity-based diversity metrics is an indirect side effect, not the primary purpose. But improvement in diversity metrics by $\alpha+\beta$ shows that β helped in fairness and diversity.

$\alpha+\beta$ = best balance

Combined model achieves highest coverage and long-tail ratio, showing both biases complement each other effectively. The small accuracy drop from β is compensated by meaningfully higher coverage and long-tail ratio

Window Size Sensitivity Analysis



W = 7

Highest NDCG@10 (short-term precision). But lowest coverage — overemphasizes trending spikes.

W = 14

Best trade-off: strong Recall@10, good coverage, good long-tail. Captures meaningful trends without noise.

W = 30

Default setting. Best coverage and long-tail ratio. Slightly lower ranking precision.

W = 60

Worst overall. Dilutes temporal dynamics, behaves like static popularity estimation.



Mean e_trend decreases consistently with larger windows. Shorter windows = more reactive but noisier trend estimates.

Limitations & Conclusion

Limitations

- ⚠ Exposure approximation: interaction frequency as proxy for true exposure (standard IPS limitation)
- ⚠ Single dataset (MovieLens-1M): limited temporal dynamics compared to industrial settings
- ⚠ Genre metadata dependency for $\alpha(u)$ — not available in all domains
- ⚠ Offline evaluation only — online A/B testing needed for full validation

Conclusion

- ✓ Personalized debiasing consistently outperforms uniform IPS on diversity and fairness metrics
- ✓ Formula-based variant excels at coverage and exposure fairness
- ✓ Learned MLP variant achieves best ranking quality (Recall@10, NDCG@10)
- ✓ Both biases (popularity + trending) complement each other for best overall balance

Thank You

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